Equity Data Audit: Cleaning, Adjustment Status, and Cross-Validation

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> *Covers 20 years of cleaned CRSP daily stock data (1995-2014), the split/dividend adjustment-status finding, and an independent cross-check against a second price file -- the output of Phase 1, Slice 2 of this project.*

What this dataset is, and why it needed cleaning

This report covers the first stage of building this project's stock-price history: cleaning a raw pull from CRSP (Center for Research in Security Prices), a widely used academic database of US daily stock prices and trading data.

The cleaned result covers 20 years (1995-2014), 20,478,337 (stock, day) rows across 8,100 distinct stocks. Each stock has a permanent numeric ID (a "permno") that never changes or gets reused, unlike its ticker symbol, which a company can change or which can be reassigned to a different company later.

Two of CRSP's own data conventions look like errors if you don't know about them, so this cleaning step checked for both directly rather than assuming they don't apply here. First, when a stock doesn't trade on a given day, CRSP records an estimated price with a minus sign in front of it as a marker that it isn't a real trade price; 110,227 rows (0.54% of the total) fall into this case, and the minus sign is removed so the price is usable, with a separate flag column kept so this is never silently hidden. Second, CRSP documents four special codes that can appear in place of a return to mean "we don't have this," not a literal huge gain or loss; this cleaned data has 0 such codes remaining.

Are prices and returns already adjusted for stock splits and dividends?

This question matters a lot for anything built on top of this data. A stock split (for example, "2-for-1") doubles the number of shares and roughly halves the price without changing what the company is actually worth or what an investor actually earned. A dividend is a cash payment to shareholders, which is part of an investor's real return even though it doesn't show up as a change in the share price. If a price series isn't adjusted for these events, a split looks exactly like a crash, and a return series that ignores dividends understates how much money was actually made.

The chart below shows one real, named example: a specific stock's actual traded price around one of its stock splits, next to what an investor's money would have looked like over that same

stretch if you build it up purely from the day-by-day return column instead of the price column. If the return column already accounts for the split, the price line should show an abrupt drop on the split date while the return-built line stays smooth straight through it -- and that is exactly what the chart shows.

Cross-checking against a second, independent price file

This project separately had access to an older, non-CRSP daily price file (one row per ticker per day) that predates its use of CRSP. Comparing that file against CRSP is a useful independent sanity check: if two differently-sourced price series agree closely, that's good evidence neither one has a scaling or adjustment mistake baked in. The two files only overlap for a few of this project's years, so this check can only speak to that overlapping stretch, not the full window this project ultimately covers.

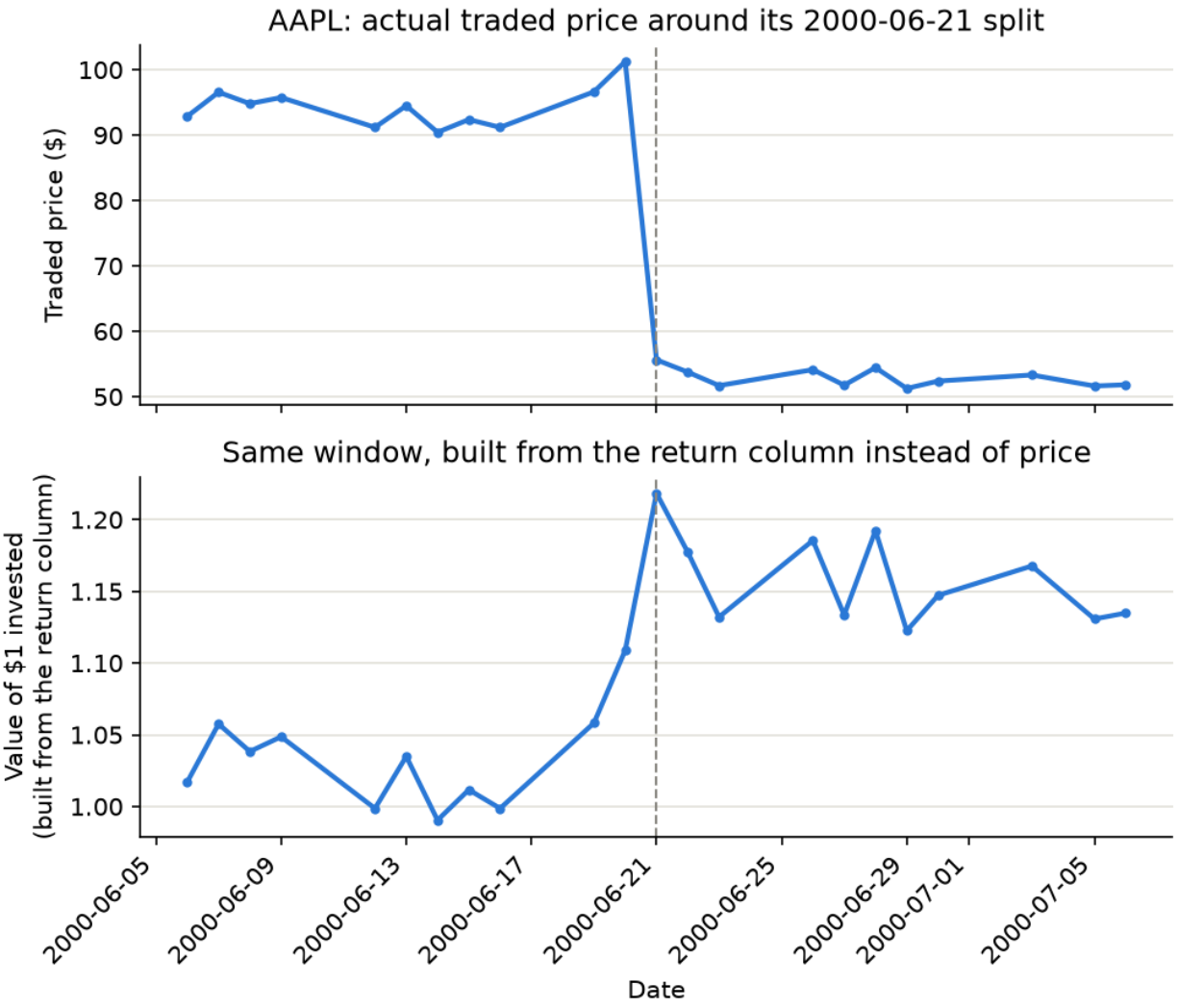
The chart below compares each matched stock-day's day-over-day return from CRSP against the same stock-day's return computed from the other file's raw prices. A point sitting exactly on the dashed diagonal line means the two sources agree perfectly on that day; the closer the whole cloud of points hugs that line, the stronger the agreement.

Coverage: how much data do we actually have?

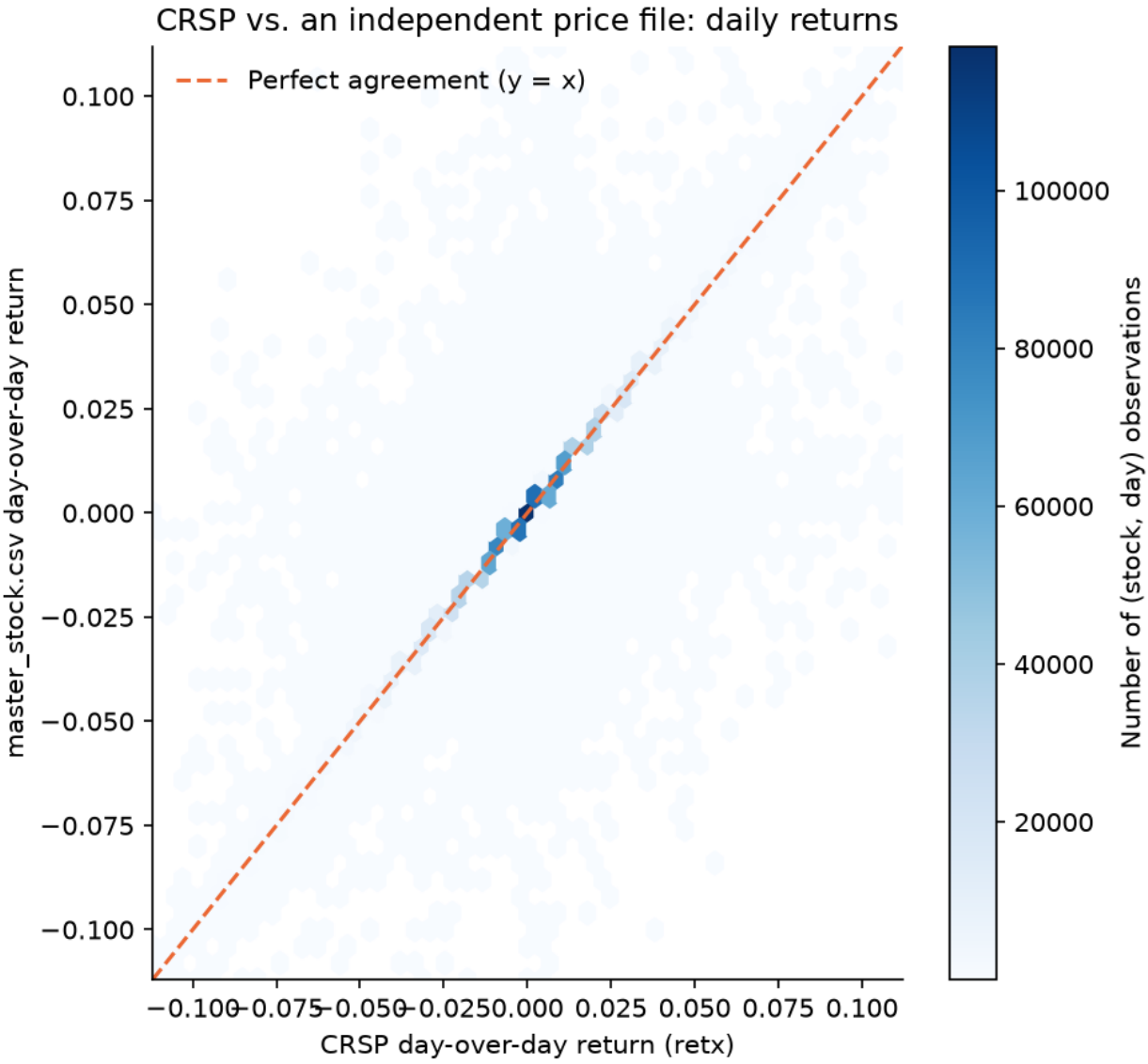
The chart below shows, for every year in the cleaned dataset, how many (stock, day) rows survived cleaning and how many distinct stocks contributed at least one of them. Coverage grows through the mid-2000s as more of the resolved stock universe has trading history on file, then narrows again as the pulled window approaches its final year.

What this means for the project

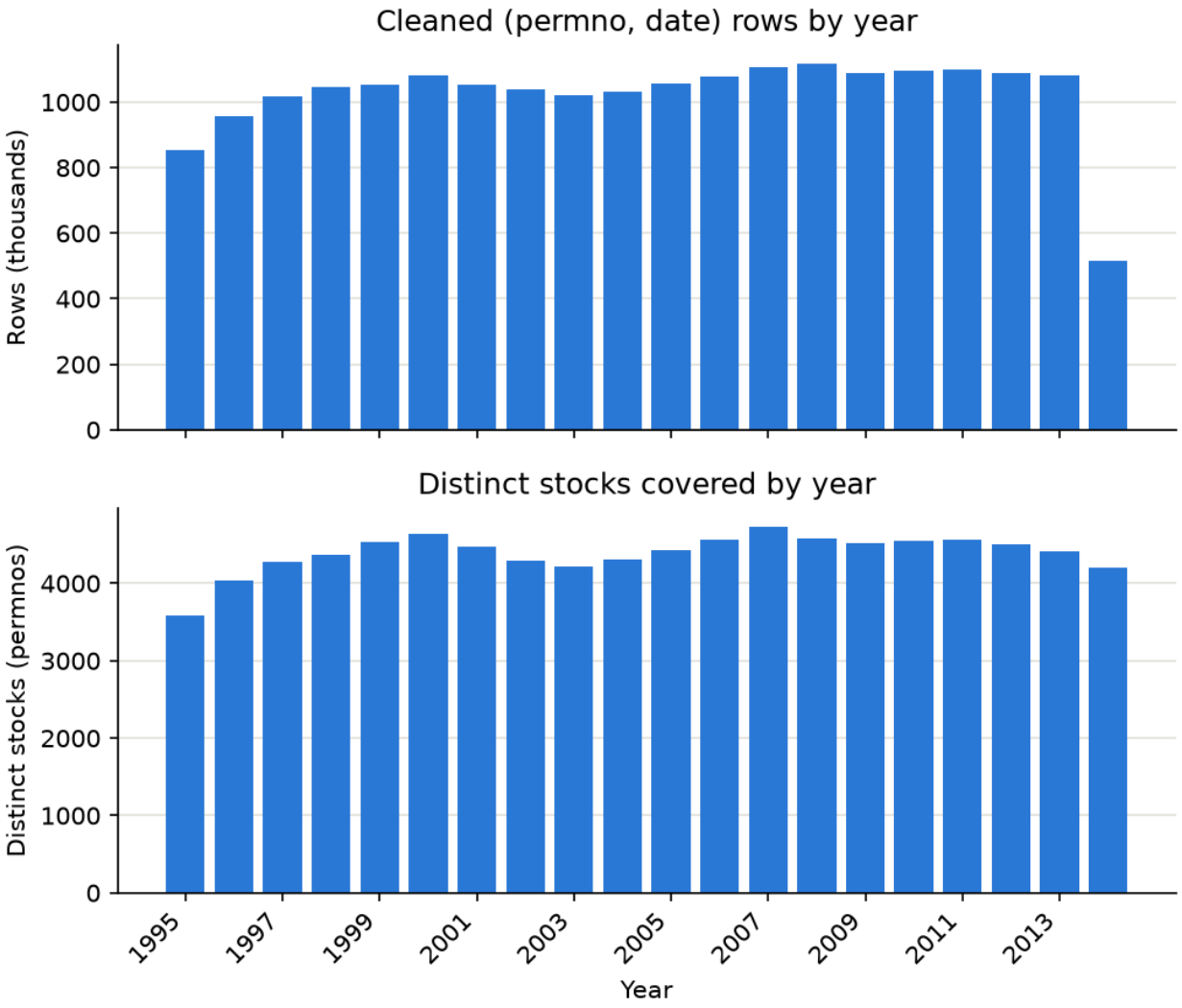
None of the checks in this report turned up a case of unadjusted returns, a missing dividend adjustment, or an implausible return scale -- this is CRSP's standard, well-documented convention, confirmed directly against this project's own pull rather than assumed. That means the cleaned return columns in data/normalized_equity/ are safe to use as-is for the return calculations later stages of this project depend on. The independent cross-check against a second price file, while it can only speak to a few of this project's years, adds further confidence: the two sources agree closely almost everywhere they overlap. The main limitation to carry forward is coverage-related, not correctness-related -- how much history exists for a given year and stock, shown in the coverage chart below.



AAPL's actual traded price fell -45.1% overnight on 2000-06-21 (from $101.25 to $55.62) purely because of the 2-for-1 split -- which would look like a huge one-day crash if you read the price column alone. The return column recorded +9.88% that day instead, and the bottom panel's return-built value of $1 invested shows no break at all: the split is already accounted for.



Each point is one (stock, day) with a return recorded in both CRSP and the independent master_stock.csv file, between 2011-01-03 and 2013-12-31 (1,250,315 stock-days shown). The two sources agree closely: a correlation of 0.9810, and 99.09% of stock-days agreeing to within 1 percentage point. Setting aside 36 ticker segments where the two files appear to be tracking two different companies under a reused ticker symbol raises agreement further, to a correlation of 0.9998 and 99.94% within 1 percentage point.



Rows and distinct stocks covered per year, 1995-2014, across 20,478,337 total rows and 8,100 distinct stocks.